Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: M5 Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
10090 Kost RD	Galt	Sacramento	95632

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 03/01/1962

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X148-X200-X013-XXXX

B. OPERATORS

Mello, Kevin

Operator name: Mello, Kevin

Telephone no.: Landline: ; Cellular: (209) 810-0066

Mailing Address Number and Street	City	State	Zip Code
10016 Kost RD	Galt	CA	95632

Mello, Lisa

Operator name: Mello, Lisa

Telephone no.: Landline: ; Cellular: (209) 810-0227

Mailing Address Number and Street	City	State	Zip Code
10016 Kost RD	Galt	CA	95632

This operator is responsible for paying permit fees.

C. OWNERS

Mello, Kevin

Legal owner name: Mello, Kevin

Telephone no.: Landline: ; Cellular: (209) 810-0066

Mailing Address Number and Street	City	State	Zip Code
10016 Kost RD	Galt	CA	95632

Mello, Lisa

Legal owner name:	Mello, Lisa	Telephone no.:		(209) 810-0227
			Landline	Cellular
10016 Kost RD		Galt	CA	95632
Mailing Address Number and Street		City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	250	75	0
Number under roof	700	100	250	0	0	120
Maximum number	700	100	250	250	75	120
Average number	700	100	250	250	75	120
Avg live weight (lbs)	1,300	1,350	1,100	750		

Predominant milk cow breed: Holstein

Average milk production: 69 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 25,074.05 *tons per reporting period*

Total nitrogen from manure: 309,356.45 *lbs per reporting period*

After ammonia losses (30% loss applied): 216,549.52 *lbs per reporting period*

Total phosphorus from manure: 50,690.61 *lbs per reporting period*

Total potassium from manure: 133,596.56 *lbs per reporting period*

Total salt from manure: 352,590.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,240,000 *gallons*

Total nitrogen generated: 43,428.72 *lbs*

Total phosphorus generated: 3,009.01 *lbs*

Total potassium generated: 80,620.71 *lbs*

Total salt generated: 540,876.17 *lbs*

	12,240,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,240,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
AG WELL #1	Ground water
AG WELL #2	Ground water
AG WELL #3	Ground water
AG WELL #4	Ground water
DOMESTIC WELL KEVINS	Ground water

Source Description	Type
DOMESTIC WELL-TRAILERS	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/20/2023	Solid commercial fertilizer *Starter Fertilizer*	3.20 *ton*	Dry-weight	8.0	16.000000	38.000000	38.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	942.08	2,237.44	2,237.44	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	942.08	2,237.44	2,237.44	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	Corral solids	2,205.00 *ton*	Dry-weight	29.3		24,200.00	3,300.00	15,500.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	75,452.45	10,288.97	48,326.99	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	75,452.45	10,288.97	48,326.99	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	28	28	2	both	X148-X200-X013-XXXX
Field 2	39	39	2	both	X148-X200-X010-XXXX
Field 3	50	50	2	both	X148-X200-X013-XXXX
Field 4	40	40	2	both	X148-X200-X013-XXXX
Field 5	90	90	4	both	X148-X200-X013-XXXX
Totals for areas that were used for application	247	247	12		
Totals for areas that were not used for application					
Land application area totals	247	247	12		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	504.00 *ton*	Dry-weight		66.4	19,100.00	3,700.00	42,700.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.00	231.03	44.76	516.50	1,391.04

Field 1

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	896.00 *ton*	Dry-weight		77.6	16,400.00	2,040.00	15,200.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	32.00	235.11	29.25	217.91	989.18

Field 2

Field name: Field 2

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	702.00 *ton*	Dry-weight		66.4	17,900.00	3,500.00	42,400.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.00	216.52	42.34	512.87	1,366.85

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,248.00 *ton*	Dry-weight		75.4	15,900.00	1,580.00	11,800.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	32.00	250.33	24.88	185.78	1,007.62

Field 3

Field name: Field 3

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	900.00 *ton*	Dry-weight		71.2	14,000.00	4,000.00	49,200.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.00	145.15	41.47	510.11	1,171.58

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,600.00 *ton*	Dry-weight		74.8	12,800.00	1,680.00	12,100.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	32.00	206.44	27.10	195.15	838.66

Field 4

Field name: Field 4

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	720.00 *ton*	Dry-weight		67.2	17,200.00	3,800.00	42,600.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.00	203.10	44.87	503.02	1,641.31

Field 4

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,280.00 *ton*	Dry-weight		76.8	16,600.00	2,360.00	16,500.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	32.00	246.48	35.04	244.99	0.00

Field 5

Field name: Field 5

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 90 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	810.00 *ton*	Dry-weight		72.8	16,600.00	3,400.00	36,500.00		11.20
05/12/2023	810.00 *ton*	Dry-weight		62.4	20,200.00	4,000.00	45,200.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	180.00	28.80	149.40	0.00
Total actual harvest content	18.00	217.99	43.72	484.62	1,306.37

Field 5

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 90 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,440.00 *ton*	Dry-weight		76.8	14,600.00	1,780.00	14,700.00		6.70
10/18/2023	1,440.00 *ton*	Dry-weight		76.2	14,900.00	1,940.00	15,100.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	32.00	256.00	48.00	211.20	0.00
Total actual harvest content	32.00	221.87	27.99	224.13	984.83

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/01/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	178.29	48.89	88.41	0.00	140.00 *ton*
Application event totals		178.29	48.89	88.41	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	51.72	3.39	100.86	510.71	360,000.00 *gal*
Application event totals		51.72	3.39	100.86	510.71	

Field 1 - 05/23/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	111.12	15.94	41.44	0.00	80.00 *ton*
Application event totals		111.12	15.94	41.44	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #1	Ground water	5.39	0.00	0.00	319.22	3,168,900.00 *gal*
Application event totals		5.39	0.00	0.00	319.22	

Field 1 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Banding	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter Fertilizer	Solid commercial fertilizer	4.00	10.00	10.00	0.00	
Application event totals		4.00	10.00	10.00	0.00	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	51.72	3.39	100.86	510.71	360,000.00 *gal*
AG WELL #1	Ground water	5.39	0.00	0.00	319.22	3,168,900.00 *gal*
Application event totals		57.11	3.39	100.86	829.94	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #1	Ground water	5.39	0.00	0.00	319.22	3,168,900.00 *gal*
Application event totals		5.39	0.00	0.00	319.22	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #1	Ground water	5.39	0.00	0.00	319.22	3,168,900.00 *gal*
Application event totals		5.39	0.00	0.00	319.22	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	73.82	5.67	123.17	1,300.39	720,000.00 *gal*
AG WELL #1	Ground water	5.39	0.00	0.00	319.13	3,168,000.00 *gal*
Application event totals		79.21	5.67	123.17	1,619.52	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #1	Ground water	5.39	0.00	0.00	319.22	3,168,900.00 *gal*
Application event totals		5.39	0.00	0.00	319.22	

Field 1 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	36.91	2.83	61.59	650.19	360,000.00 *gal*
AG WELL #1	Ground water	5.39	0.00	0.00	319.22	3,168,900.00 *gal*
Application event totals		42.30	2.83	61.59	969.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #1	Ground water	5.39	0.00	0.00	319.22	3,168,900.00 *gal*
Application event totals		5.39	0.00	0.00	319.22	

Field 2 - 11/01/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	182.86	50.14	90.67	0.00	200.00 *ton*
Application event totals		182.86	50.14	90.67	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	74.26	4.87	144.82	733.33	720,000.00 *gal*
Application event totals		74.26	4.87	144.82	733.33	

Field 2 - 05/23/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage Plant date: 05/23/2023

Field 2 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	99.73	14.31	37.19	0.00	100.00 *ton*
Application event totals		99.73	14.31	37.19	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	7.43	0.00	0.00	366.95	4,224,000.00 *gal*
Application event totals		7.43	0.00	0.00	366.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Banding	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter Fertilizer	Solid commercial fertilizer	4.00	10.00	10.00	0.00	
Application event totals		4.00	10.00	10.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	7.43	0.00	0.00	366.95	4,224,000.00 *gal*
Application event totals		7.43	0.00	0.00	366.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	74.26	4.87	144.82	733.33	720,000.00 *gal*
AG WELL #2	Ground water	7.43	0.00	0.00	366.95	4,224,000.00 *gal*
Application event totals		81.69	4.87	144.82	1,100.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	7.43	0.00	0.00	366.95	4,224,000.00 *gal*
Application event totals		7.43	0.00	0.00	366.95	

Field 2 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	53.00	4.07	88.43	933.61	720,000.00 *gal*
AG WELL #2	Ground water	7.43	0.00	0.00	366.95	4,224,000.00 *gal*
Application event totals		60.43	4.07	88.43	1,300.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	7.43	0.00	0.00	366.95	4,224,000.00 *gal*
Application event totals		7.43	0.00	0.00	366.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	53.00	4.07	88.43	933.61	720,000.00 *gal*
AG WELL #2	Ground water	7.43	0.00	0.00	366.95	4,224,000.00 *gal*
Application event totals		60.43	4.07	88.43	1,300.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	7.43	0.00	0.00	366.95	4,224,000.00 *gal*
Application event totals		7.43	0.00	0.00	366.95	

Field 3 - 11/01/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	164.02	44.98	81.34	0.00	230.00 *ton*
Application event totals		164.02	44.98	81.34	0.00	

04/08/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	28.96	1.90	56.48	286.00	360,000.00 *gal*
Application event totals		28.96	1.90	56.48	286.00	

Field 3 - 05/23/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	93.34	13.39	34.81	0.00	120.00 *ton*
Application event totals		93.34	13.39	34.81	0.00	

05/23/2023	Banding	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter Fertilizer	Solid commercial fertilizer	4.00	10.00	10.00	0.00	
Application event totals		4.00	10.00	10.00	0.00	

05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.21	0.00	0.00	208.16	3,072,000.00 *gal*
AG WELL #3	Ground water	9.08	0.00	0.00	257.38	2,668,000.00 *gal*
Application event totals		13.30	0.00	0.00	465.54	

Field 3 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	57.92	3.80	112.96	572.00	720,000.00 *gal*
AG WELL #2	Ground water	4.21	0.00	0.00	208.16	3,072,000.00 *gal*
AG WELL #3	Ground water	9.08	0.00	0.00	257.38	2,668,000.00 *gal*
Application event totals		71.22	3.80	112.96	1,037.54	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.21	0.00	0.00	208.16	3,072,000.00 *gal*
AG WELL #3	Ground water	9.08	0.00	0.00	257.38	2,668,000.00 *gal*
Application event totals		13.30	0.00	0.00	465.54	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.21	0.00	0.00	208.16	3,072,000.00 *gal*
AG WELL #3	Ground water	9.08	0.00	0.00	257.38	2,668,000.00 *gal*
Application event totals		13.30	0.00	0.00	465.54	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	20.67	1.59	34.49	364.11	360,000.00 *gal*
AG WELL #2	Ground water	4.21	0.00	0.00	208.16	3,072,000.00 *gal*
AG WELL #3	Ground water	9.08	0.00	0.00	257.38	2,668,000.00 *gal*
Application event totals		33.97	1.59	34.49	829.65	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.21	0.00	0.00	208.16	3,072,000.00 *gal*
AG WELL #3	Ground water	9.08	0.00	0.00	257.38	2,668,000.00 *gal*
Application event totals		13.30	0.00	0.00	465.54	

Field 3 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.21	0.00	0.00	208.16	3,072,000.00 *gal*
AG WELL #3	Ground water	9.08	0.00	0.00	257.38	2,668,000.00 *gal*
Application event totals		13.30	0.00	0.00	465.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.21	0.00	0.00	208.16	3,072,000.00 *gal*
AG WELL #3	Ground water	9.08	0.00	0.00	257.38	2,668,000.00 *gal*
Application event totals		13.30	0.00	0.00	465.54	

Field 4 - 11/01/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	178.29	48.89	88.41	0.00	200.00 *ton*
Application event totals		178.29	48.89	88.41	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	72.40	4.75	141.20	715.00	720,000.00 *gal*
Application event totals		72.40	4.75	141.20	715.00	

Field 4 - 05/23/2023: Corn, silage

Field 4 - 05/23/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	97.23	13.95	36.26	0.00	100.00 *ton*
Application event totals		97.23	13.95	36.26	0.00	

05/23/2023	Banding	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter Fertilizer	Solid commercial fertilizer	4.00	10.00	10.00	0.00	
Application event totals		4.00	10.00	10.00	0.00	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.28	0.00	0.00	211.42	2,496,000.00 *gal*
AG WELL #3	Ground water	9.29	0.00	0.00	263.36	2,184,000.00 *gal*
Application event totals		13.58	0.00	0.00	474.77	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	72.40	4.75	141.20	715.00	720,000.00 *gal*
AG WELL #2	Ground water	4.28	0.00	0.00	211.42	2,496,000.00 *gal*
AG WELL #3	Ground water	9.29	0.00	0.00	263.36	2,184,000.00 *gal*
Application event totals		85.98	4.75	141.20	1,189.77	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.28	0.00	0.00	211.42	2,496,000.00 *gal*
AG WELL #3	Ground water	9.29	0.00	0.00	263.36	2,184,000.00 *gal*
Application event totals		13.58	0.00	0.00	474.77	

Field 4 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	25.84	1.98	43.11	455.14	360,000.00 *gal*
AG WELL #2	Ground water	4.28	0.00	0.00	211.42	2,496,000.00 *gal*
AG WELL #3	Ground water	9.29	0.00	0.00	263.36	2,184,000.00 *gal*
Application event totals		39.41	1.98	43.11	929.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.28	0.00	0.00	211.42	2,496,000.00 *gal*
AG WELL #3	Ground water	9.29	0.00	0.00	263.36	2,184,000.00 *gal*
Application event totals		13.58	0.00	0.00	474.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	25.84	1.98	43.11	455.14	360,000.00 *gal*
AG WELL #2	Ground water	6.00	0.00	0.00	296.12	3,496,000.00 *gal*
AG WELL #3	Ground water	9.29	0.00	0.00	263.36	2,184,000.00 *gal*
Application event totals		41.13	1.98	43.11	1,014.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.28	0.00	0.00	211.42	2,496,000.00 *gal*
AG WELL #3	Ground water	9.29	0.00	0.00	263.36	2,184,000.00 *gal*
Application event totals		13.58	0.00	0.00	474.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #2	Ground water	4.28	0.00	0.00	211.42	2,496,000.00 *gal*
AG WELL #3	Ground water	9.29	0.00	0.00	263.36	2,184,000.00 *gal*
Application event totals		13.58	0.00	0.00	474.77	

Field 5 - 11/01/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	178.29	48.89	88.41	0.00	450.00 *ton*
Application event totals		178.29	48.89	88.41	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	64.36	4.22	125.51	635.56	1,440,000.00 *gal*
Application event totals		64.36	4.22	125.51	635.56	

Field 5 - 05/23/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MANURE APPLICATION	Scraped material	108.04	15.50	40.29	0.00	250.00 *ton*
Application event totals		108.04	15.50	40.29	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Banding	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter Fertilizer	Solid commercial fertilizer	4.00	10.00	10.00	0.00	
Application event totals		4.00	10.00	10.00	0.00	

Field 5 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #3	Ground water	9.22	0.00	0.00	261.11	4,872,000.00 *gal*
AG WELL #4	Ground water	0.71	0.00	0.00	95.77	4,872,000.00 *gal*
Application event totals		9.92	0.00	0.00	356.88	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	48.27	3.16	94.13	476.67	1,080,000.00 *gal*
AG WELL #3	Ground water	9.22	0.00	0.00	261.11	4,872,000.00 *gal*
AG WELL #4	Ground water	0.71	0.00	0.00	95.77	4,872,000.00 *gal*
Application event totals		58.19	3.16	94.13	833.54	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #3	Ground water	9.22	0.00	0.00	261.11	4,872,000.00 *gal*
AG WELL #4	Ground water	0.71	0.00	0.00	95.77	4,872,000.00 *gal*
Application event totals		9.92	0.00	0.00	356.88	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #3	Ground water	9.22	0.00	0.00	261.11	4,872,000.00 *gal*
AG WELL #4	Ground water	0.71	0.00	0.00	95.77	4,872,000.00 *gal*
Application event totals		9.92	0.00	0.00	356.88	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	22.97	1.76	38.32	404.57	720,000.00 *gal*
AG WELL #3	Ground water	9.22	0.00	0.00	261.11	4,872,000.00 *gal*
AG WELL #4	Ground water	0.71	0.00	0.00	95.77	4,872,000.00 *gal*
Application event totals		32.89	1.76	38.32	761.44	

Field 5 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #3	Ground water	9.22	0.00	0.00	261.11	4,872,000.00 *gal*
AG WELL #4	Ground water	0.71	0.00	0.00	95.77	4,872,000.00 *gal*
Application event totals		9.92	0.00	0.00	356.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	22.97	1.76	38.32	404.57	720,000.00 *gal*
AG WELL #3	Ground water	9.22	0.00	0.00	261.11	4,872,000.00 *gal*
AG WELL #4	Ground water	0.71	0.00	0.00	95.77	4,872,000.00 *gal*
Application event totals		32.89	1.76	38.32	761.44	

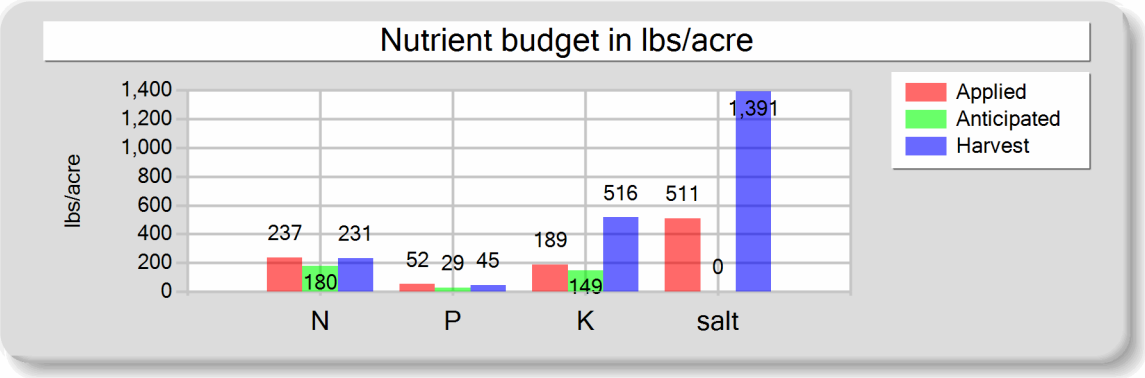
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
AG WELL #3	Ground water	9.22	0.00	0.00	261.11	4,872,000.00 *gal*
AG WELL #4	Ground water	0.71	0.00	0.00	95.77	4,872,000.00 *gal*
Application event totals		9.92	0.00	0.00	356.88	

B. NUTRIENT BUDGET

Field 1 - 11/01/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.29	48.89	88.41	0.00
Process wastewater	51.72	3.39	100.86	510.71
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.00	52.28	189.26	510.71
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	231.03	44.76	516.50	1,391.04
Nutrient balance	5.97	7.53	-327.24	-880.33
Applied to removed ratio	1.03	1.17	0.37	0.37

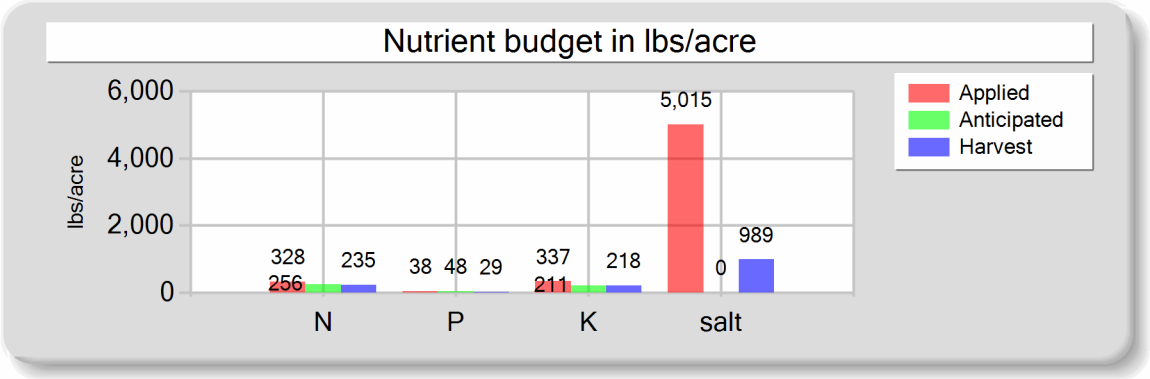
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
360,000.00	*gallons*
13.26	*acre-inches*
0.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/23/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	4.00	10.00	10.00	0.00
Dry manure	111.12	15.94	41.44	0.00
Process wastewater	162.44	11.89	285.61	2,461.30
Fresh water	43.14	0.00	0.00	2,553.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	327.71	37.83	337.06	5,014.99
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	235.11	29.25	217.91	989.18
Nutrient balance	92.60	8.58	119.15	4,025.80
Applied to removed ratio	1.39	1.29	1.55	5.07

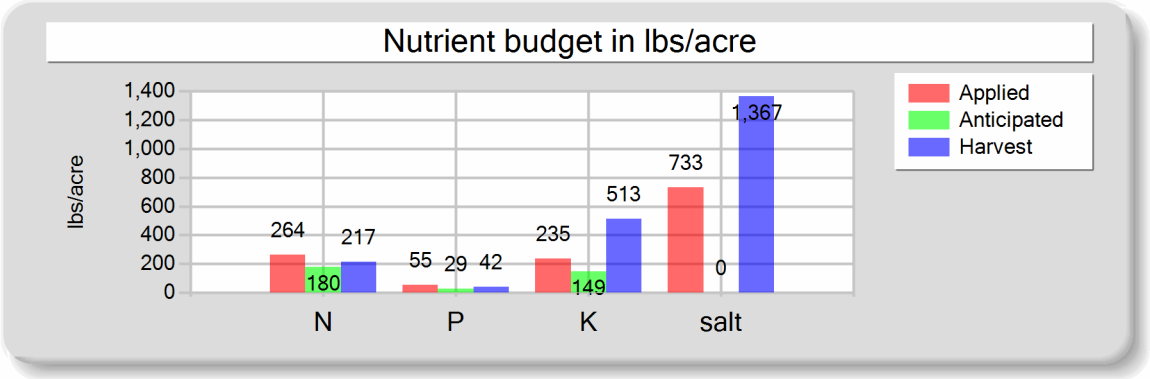
Fresh water applied	
25,350,300.00	*gallons*
933.57	*acre-inches*
33.34	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/01/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	182.86	50.14	90.67	0.00
Process wastewater	74.26	4.87	144.82	733.33
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	264.12	55.01	235.49	733.33
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	216.52	42.34	512.87	1,366.85
Nutrient balance	47.60	12.68	-277.38	-633.52
Applied to removed ratio	1.22	1.30	0.46	0.54

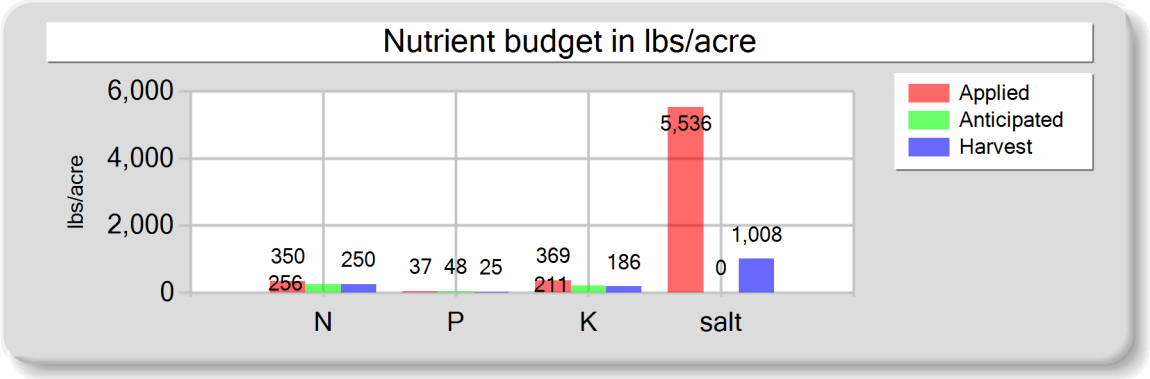
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
720,000.00	*gallons*
26.52	*acre-inches*
0.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/23/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	4.00	10.00	10.00	0.00
Dry manure	99.73	14.31	37.19	0.00
Process wastewater	180.25	13.00	321.68	2,600.56
Fresh water	59.44	0.00	0.00	2,935.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	350.41	37.31	368.87	5,536.19
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	250.33	24.88	185.78	1,007.62
Nutrient balance	100.09	12.43	183.09	4,528.58
Applied to removed ratio	1.40	1.50	1.99	5.49

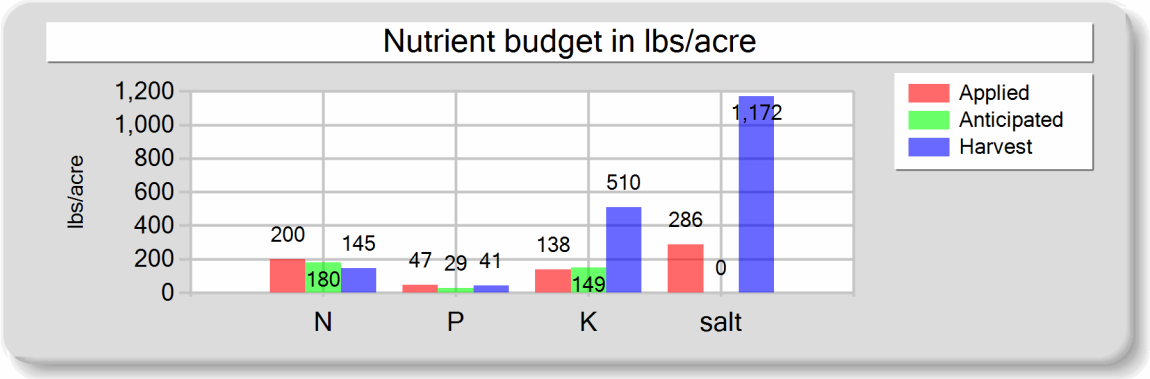
Fresh water applied	
33,792,000.00	*gallons*
1,244.44	*acre-inches*
31.91	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
2.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/01/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.02	44.98	81.34	0.00
Process wastewater	28.96	1.90	56.48	286.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	199.98	46.88	137.81	286.00
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	145.15	41.47	510.11	1,171.58
Nutrient balance	54.83	5.41	-372.29	-885.58
Applied to removed ratio	1.38	1.13	0.27	0.24

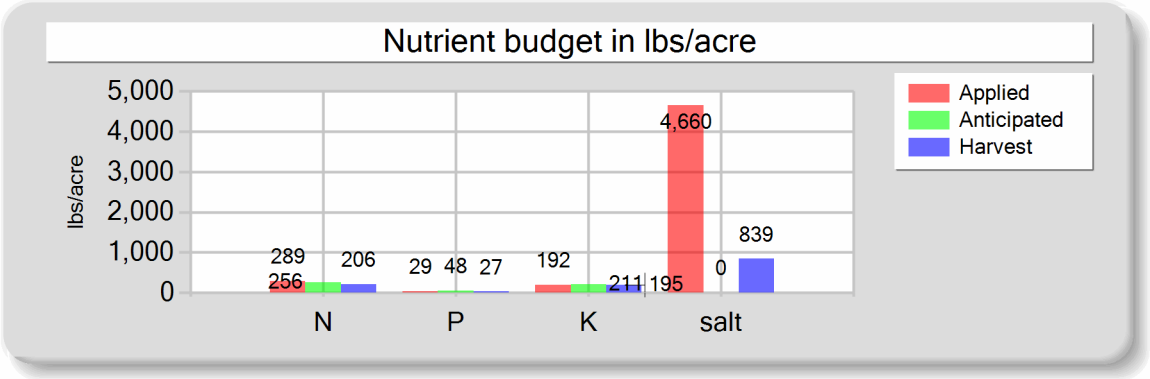
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
360,000.00 *gallons*
13.26 *acre-inches*
0.27 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 3 - 05/23/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	4.00	10.00	10.00	0.00
Dry manure	93.34	13.39	34.81	0.00
Process wastewater	78.59	5.38	147.45	936.11
Fresh water	106.39	0.00	0.00	3,724.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.32	28.77	192.26	4,660.43
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	206.44	27.10	195.15	838.66
Nutrient balance	82.88	1.68	-2.89	3,821.77
Applied to removed ratio	1.40	1.06	0.99	5.56

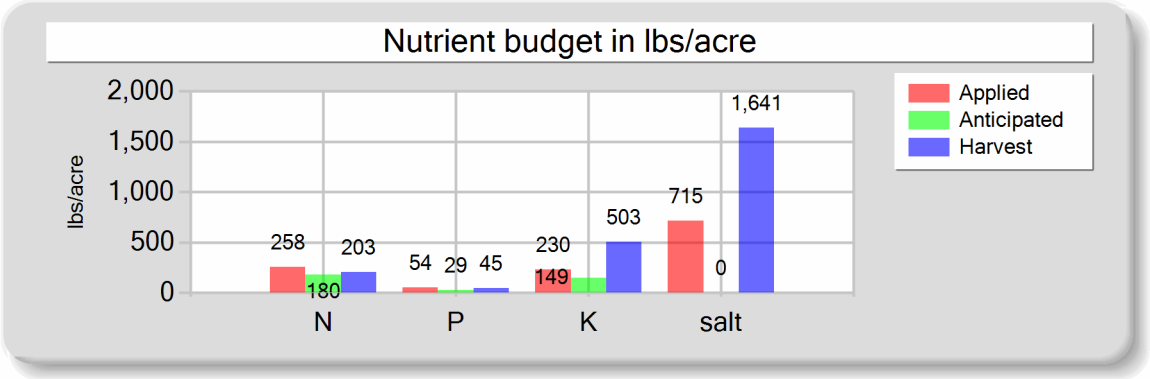
Fresh water applied	
45,920,000.00	*gallons*
1,691.08	*acre-inches*
33.82	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
0.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/01/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.29	48.89	88.41	0.00
Process wastewater	72.40	4.75	141.20	715.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	257.69	53.64	229.61	715.00
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	203.10	44.87	503.02	1,641.31
Nutrient balance	54.59	8.77	-273.42	-926.31
Applied to removed ratio	1.27	1.20	0.46	0.44

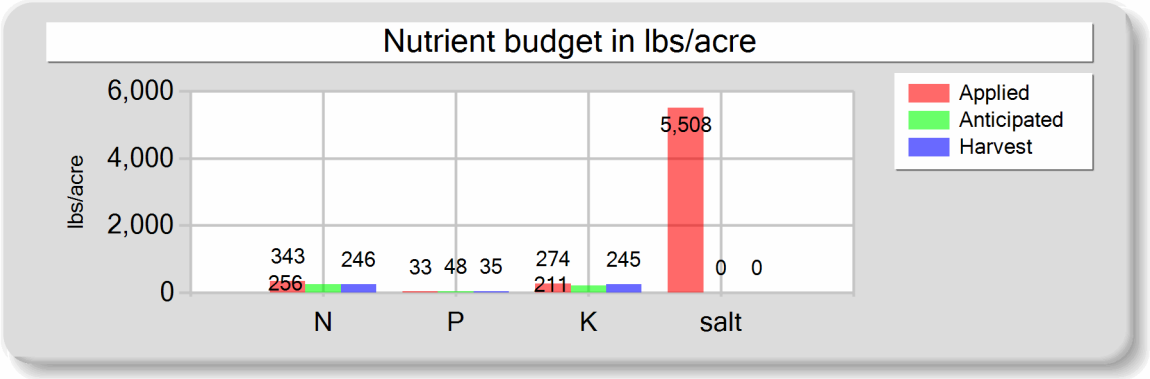
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
720,000.00 *gallons*
26.52 *acre-inches*
0.66 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 4 - 05/23/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	4.00	10.00	10.00	0.00
Dry manure	97.23	13.95	36.26	0.00
Process wastewater	124.07	8.71	227.42	1,625.27
Fresh water	110.32	0.00	0.00	3,882.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	342.63	32.66	273.68	5,508.16
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	246.48	35.04	244.99	0.00
Nutrient balance	96.15	-2.38	28.69	5,508.16
Applied to removed ratio	1.39	0.93	1.12	0.00

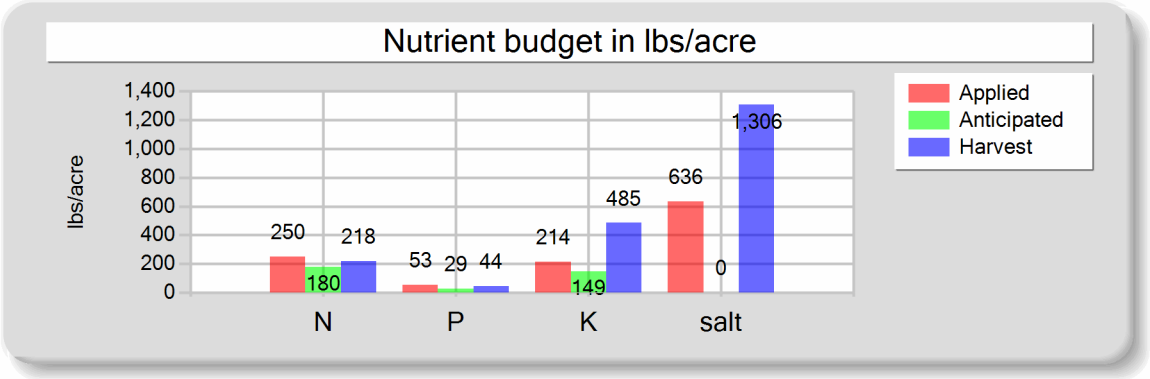
Fresh water applied	
38,440,000.00	*gallons*
1,415.61	*acre-inches*
35.39	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/01/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.29	48.89	88.41	0.00
Process wastewater	64.36	4.22	125.51	635.56
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	249.64	53.11	213.92	635.56
Anticipated crop nutrient removal	180.00	28.80	149.40	0.00
Actual crop nutrient removal	217.99	43.72	484.62	1,306.37
Nutrient balance	31.66	9.39	-270.70	-670.81
Applied to removed ratio	1.15	1.21	0.44	0.49

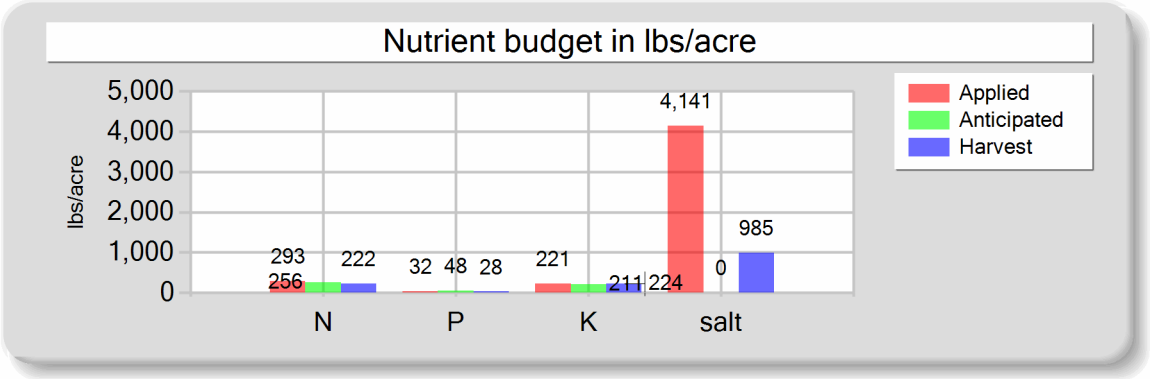
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,440,000.00 *gallons*
53.03 *acre-inches*
0.59 *inches/acre*

Total harvests for the crop
2 *harvests*

Field 5 - 05/23/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	4.00	10.00	10.00	0.00
Dry manure	108.04	15.50	40.29	0.00
Process wastewater	94.20	6.69	170.77	1,285.80
Fresh water	79.40	0.00	0.00	2,855.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.63	32.19	221.06	4,140.81
Anticipated crop nutrient removal	256.00	48.00	211.20	0.00
Actual crop nutrient removal	221.87	27.99	224.13	984.83
Nutrient balance	70.77	4.20	-3.07	3,155.98
Applied to removed ratio	1.32	1.15	0.99	4.20

Fresh water applied	
77,952,000.00	*gallons*
2,870.71	*acre-inches*
31.90	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
1.03	*inches/acre*

Total harvests for the crop	
2	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2287057A

Sample and source description: T2287057A

Sample date: 10/14/2022 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,400.00	5,320.00	9,620.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3132019B

Sample and source description: T3132019B

Sample date: 05/12/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 20.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,400.00	3,500.00	9,100.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3254026B

Sample and source description: T3254026B

Sample date: 09/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	3,800.00	12,600.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3291062A

Sample and source description: T3291062A

Sample date: 10/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 29.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	3,300.00	15,500.00	0.00	0.00	0.00	0.00	0.00		0.00
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3132019A

Sample and source description: T3132019A

Sample date: 05/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	482.00	319.00	0.00	0.00	31.60	940.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	705.00	4,760
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

T3254026A

Sample and source description: T3254026A

Sample date: 09/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	344.00	269.00	0.00	0.00	26.40	574.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	670.00	6,060
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

T3291062B

Sample and source description: T3291062B

Sample date: 10/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	468.00	294.00	0.00	0.00	29.20	880.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	676.00	5,720
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

C. FRESH WATER ANALYSES

AG WELL #1

T3254027A

Sample description: T3254027A

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.71	0.00	5.71	0.00	0.00	0.00	0.00	0.00	0.00	0.00	528.00	338
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

AG WELL #2

T3254027B

Sample description: T3254027B

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.22	0.00	8.22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	635.00	406
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

AG WELL #3

AG WELL #3

T3254027D

Sample description: T3254027D

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	20.40	0.00	20.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	903.00	578
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

AG WELL #4

T3254027E

Sample description: T3254027E

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.57	0.00	1.57	0.00	0.00	0.00	0.00	0.00	0.00	0.00	331.00	212
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

DOMESTIC WELL KEVINS

T3254027F

Sample description: T3254027F

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.77	0.00	1.77	0.00	0.00	0.00	0.00	0.00	0.00	0.00	333.00	213
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

DOMESTIC WELL-TRAILERS

DOMESTIC WELL-TRAILERS

T3254027C

Sample description: T3254027C

Sample date: 09/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.48	0.00	8.48	0.00	0.00	0.00	0.00	0.00	0.00	0.00	773.00	494
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 11/01/2022: Oats, silage-soft dough

T3132020A

Sample and source description: T3132020A

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	3,700.00	42,700.00		11.50
DL	1.00	1.00	1.00		1.00

Field 1 - 05/23/2023: Corn, silage

Field 1 - 05/23/2023: Corn, silage

T3291063A

Sample and source description: T3291063A

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	2,040.00	15,200.00		6.90
DL	1.00	1.00	1.00		1.00

Field 2 - 11/01/2022: Oats, silage-soft dough

T3132020B

Sample and source description: T3132020B

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,900.00	3,500.00	42,400.00		11.30
DL	1.00	1.00	1.00		1.00

Field 2 - 05/23/2023: Corn, silage

T3291063B

Sample and source description: T3291063B

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	1,580.00	11,800.00		6.40
DL	1.00	1.00	1.00		1.00

Field 3 - 11/01/2022: Oats, silage-soft dough

T3132020C

Sample and source description: T3132020C

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	4,000.00	49,200.00		11.30
DL	1.00	1.00	1.00		1.00

Field 3 - 05/23/2023: Corn, silage

T3291063C

Sample and source description: T3291063C

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	1,680.00	12,100.00		5.20
DL	1.00	1.00	1.00		1.00

Field 4 - 11/01/2022: Oats, silage-soft dough

T3132020D

Sample and source description: T3132020D

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,800.00	42,600.00		13.90
DL	1.00	1.00	1.00		1.00

Field 4 - 05/23/2023: Corn, silage

T3291063D

Sample and source description: T3291063D

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,600.00	2,360.00	16,500.00		6.80
DL	1.00	1.00	1.00		1.00

Field 5 - 11/01/2022: Oats, silage-soft dough

T3132020E

Sample and source description: T3132020E

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,600.00	3,400.00	36,500.00		11.20
DL	1.00	1.00	1.00		1.00

T3132020F

Sample and source description: T3132020F

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,200.00	4,000.00	45,200.00		11.20
DL	1.00	1.00	1.00		1.00

Field 5 - 05/23/2023: Corn, silage

Field 5 - 05/23/2023: Corn, silage

T3291063E

Sample and source description: T3291063E

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	1,780.00	14,700.00		6.70
DL	1.00	1.00	1.00		1.00

T3291063F

Sample and source description: T3291063F

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	1,940.00	15,100.00		6.40
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

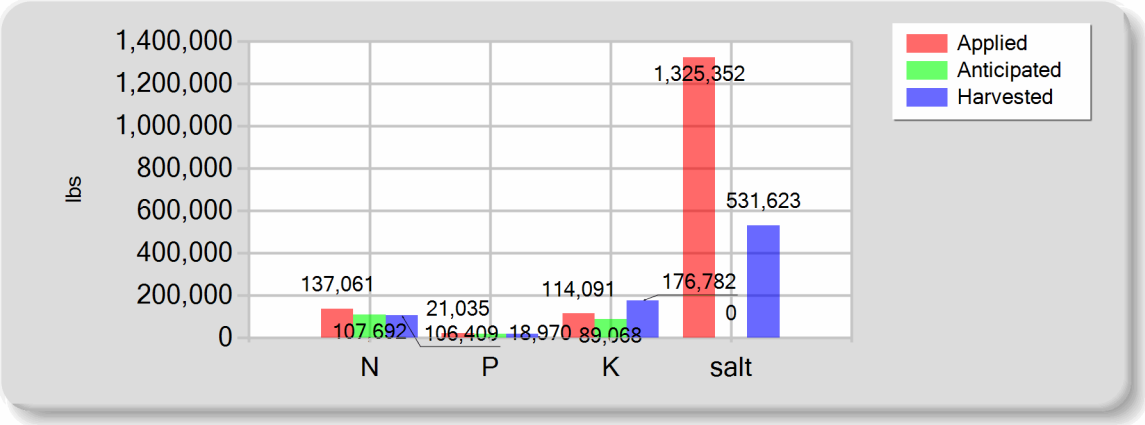
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

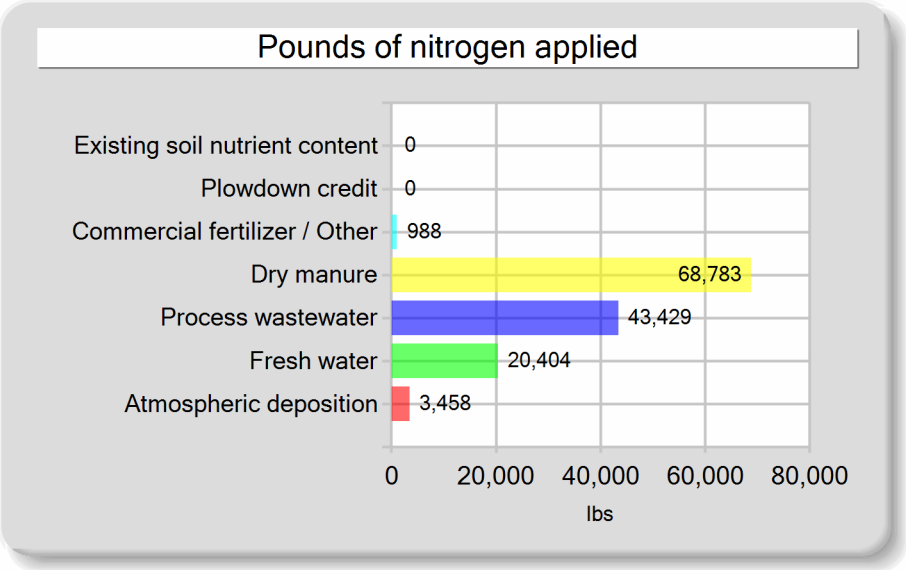
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

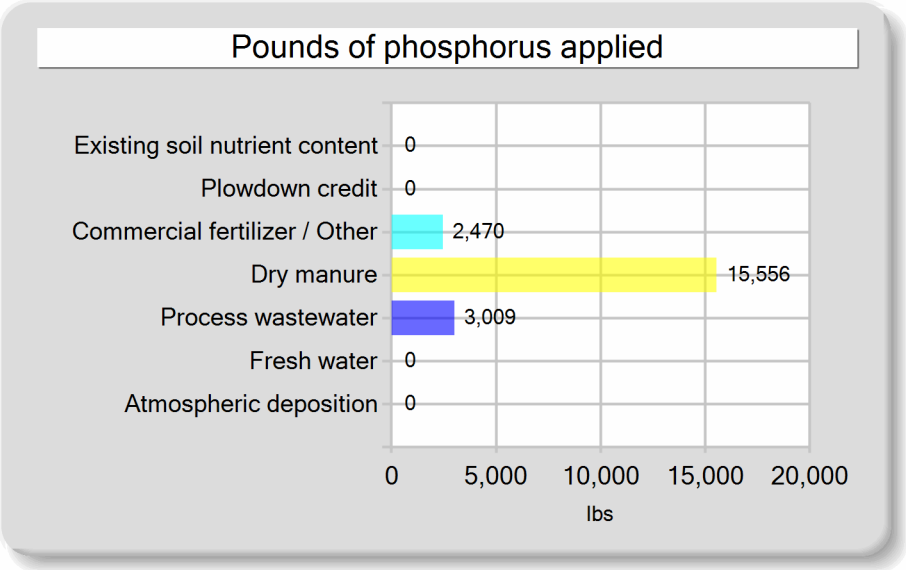
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	988.00	2,470.00	2,470.00	0.00
Dry manure	68,782.62	15,555.71	31,000.01	0.00
Process wastewater	43,428.72	3,009.01	80,620.71	540,876.17
Fresh water	20,403.87	0.00	0.00	784,475.94
Atmospheric deposition	3,458.00	0.00	0.00	0.00
Total nutrients applied	137,061.21	21,034.71	114,090.72	1,325,352.11
Anticipated crop nutrient removal	107,692.00	18,969.60	89,068.20	0.00
Actual crop nutrient removal	106,408.64	17,771.82	176,781.63	531,622.85
Nutrient balance	30,652.57	3,262.89	-62,690.90	793,729.26
Applied to removed ratio	1.29	1.18	0.65	2.49

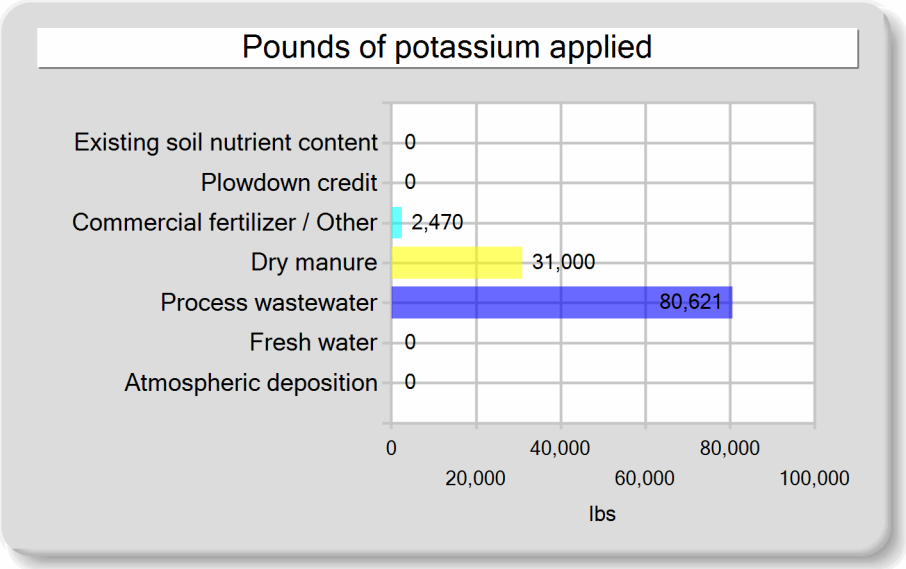
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

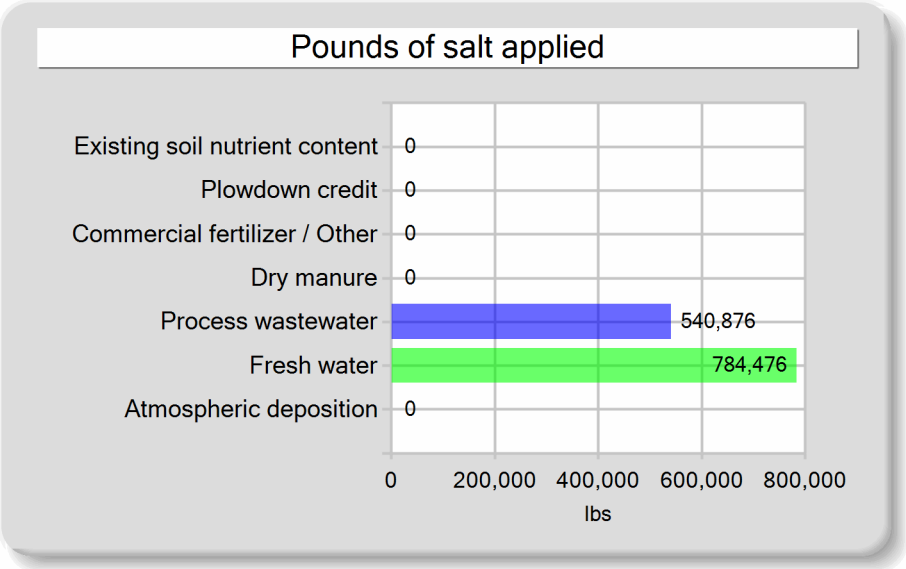


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Lisa Mello

SIGNATURE OF OWNER OF FACILITY

Lisa Mello

PRINT OR TYPE NAME

6-27-24

DATE

Kevin Mello

SIGNATURE OF OPERATOR OF FACILITY

Kevin Mello

PRINT OR TYPE NAME

6-27-24

DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Kevin Mello

Name of Dairy Facility: M5 Dairy

Facility Address: 10090 Kost Rd, Galt, CA. 95632
Number and Street, City, Zip Code

Contact Person Name and Phone Number: Kevin Mello 209-745-5017
Name, Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Witt Ranch Transport

Address of Hauling Company /Person: P.O. Box 428, Lodi, CA. 95241
Number and Street, City, Zip Code

Contact Person: Fred Witt 209-483-9542
Name, Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

L.V. Thompson 21030 N. Davis Rd, Lodi CA. 95242
Name, Number and Street, City, Zip Code, Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

23030 N. Davis Rd, Lodi, CA 95242
Number and Street, City, Zip Code, Assessor's Parcel Number 01308007

Dates Hauled: 10-3-23 - 10-24-23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: (45 yards per Truck Load) (1575 Cubic Yards)

Manure: 2,205 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: 35 Loads

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: N/A Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes X No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after 31 December 2007 to such party.
N/A (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Kevin Mellor Date: 11-14-23

Hauler's Signature: Ted Witt Date: 11-14-23

DENELE ANALYTICAL, INC.

1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: MS Dairy / Kevin + Lisa Mello
Mailing address: 10016 Kost Rd.
(not facility address) Galt, CA. 95632
Phone#:

Authorized Copy Released To:

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

*Required Field Measurement

___ (Check here if perfomed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	
							Analysis				
1	9-11-23	7:40am	Ag Well # 1		23.8		X				
2	9-11-23	7:50am	Ag Well # 2		23.6		X				
3	9-11-23	8:00am	Domestic Well-	Trailers	23.8		X				
4	9-11-23	8:19am	Ag Well #3		23.0		✓				
5	9-11-23	8:32am	Ag Well #4		22.8		X				
6	9-11-23	8:50am	Domestic Well-	Kevin's	23.8		X				
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY)	
1st	Kevin Mello	MS Dairy		9-11-23 1:42pm	Logged in By:	LABORATORY NUMBER
2nd		DENELE ANALYTICAL, INC.	9-11-23 1:42			T3-54027

Jun.29.2018 09:20 Denele Labs: Woodland 5306669058

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
M5 Dairy
10016 Kost Rd.
Galt, CA 95632

Grower

Consultant:

PURCHASE ORDER: Kevin & Lisa Mello
RECEIVED DATE: 9/11/23 1:42 PM
SUBMITTED BY: Kevin
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 23.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well #1	T3254027A	Electrical Conductivity (EC)	EPA 120.1 and	0.528	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.71	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	338	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	98.9			0.158
Nitrate Nitrogen (NO3-N)	97.1	92.8 (T3254019E)	91.2 (T3254019E)	1.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
M5 Dairy
10016 Kost Rd.
Galt, CA 95632

Grower

Consultant:

PURCHASE ORDER: Kevin & Lisa Mello
RECEIVED DATE: 9/11/23 1:42 PM
SUBMITTED BY: Kevin
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 23.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well #2	T3254027B	Electrical Conductivity (EC)	EPA 120.1 and	0.635	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.22	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	406	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	98.9			0.158
Nitrate Nitrogen (NO3-N)	97.1	92.8 (T3254019E)	91.2 (T3254019E)	1.65

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Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
M5 Dairy
10016 Kost Rd.
Galt, CA 95632

Grower

Consultant:

PURCHASE ORDER: Kevin & Lisa Mello
RECEIVED DATE: 9/11/23 1:42 PM
SUBMITTED BY: Kevin
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 23.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well Trailers	T3254027C	Electrical Conductivity (EC)	EPA 120.1 and	0.773	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.48	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	494	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	98.9			0.158
Nitrate Nitrogen (NO3-N)	97.1	92.8 (T3254019E)	91.2 (T3254019E)	1.65

e warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the tomer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
M5 Dairy
10016 Kost Rd.
Galt, CA 95632

Grower

Consultant:

PURCHASE ORDER: Kevin & Lisa Mello
RECEIVED DATE: 9/11/23 1:42 PM
SUBMITTED BY: Kevin
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 23.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well #3	T3254027D	Electrical Conductivity (EC)	EPA 120.1 and	0.903	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	20.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	578	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	98.9			0.158
Nitrate Nitrogen (NO3-N)	97.1	92.8 (T3254019E)	91.2 (T3254019E)	1.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
M5 Dairy
10016 Kost Rd.
Galt, CA 95632

Grower

Consultant:

PURCHASE ORDER: Kevin & Lisa Mello
RECEIVED DATE: 9/11/23 1:42 PM
SUBMITTED BY: Kevin
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 22.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well #4	T3254027E	Electrical Conductivity (EC)	EPA 120.1 and	0.331	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.57	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	212	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	98.9			0.158
Nitrate Nitrogen (NO3-N)	97.1	92.8 (T3254019E)	91.2 (T3254019E)	1.65

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Compliance Analysis Report

Customer
M5 Dairy
10016 Kost Rd.
Galt, CA 95632

Grower

Consultant:

PURCHASE ORDER: Kevin & Lisa Mello
RECEIVED DATE: 9/11/23 1:42 PM
SUBMITTED BY: Kevin
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 23.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well Kevins	T3254027F	Electrical Conductivity (EC)	EPA 120.1 and	0.333	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.77	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	213	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.6	95.8 (T3249019A)	93.4 (T3249019A)	2.53
Electrical Conductivity (EC)	98.9			0.158
Nitrate Nitrogen (NO3-N)	97.1	92.8 (T3254019E)	91.2 (T3254019E)	1.65

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